

Bitcoin Market Sentiment and Trader Performance Analysis

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1 Introduction

Cryptocurrency markets are highly influenced by investor sentiment. The Fear and Greed Index is widely used to measure market emotions and classify market conditions into categories such as Extreme Fear, Fear, Neutral, Greed, and Extreme Greed.

The objective of this project is to analyze the relationship between Bitcoin market sentiment and trader performance using historical trading data from Hyperliquid and the Bitcoin Fear and Greed Index dataset.

2 Objectives

- Analyze trader performance under different market sentiments.
- Measure average profit and total profit across sentiment categories.
- Calculate trader win rates.
- Identify patterns that can help in building better trading strategies.

3 Datasets Used

3.1 Historical Trader Data

The dataset contains historical trade information including:

- Account
- Coin
- Execution Price
- Size Tokens
- Side
- Timestamp
- Closed PnL
- Fees

3.2 Bitcoin Fear and Greed Index

The sentiment dataset contains:

- Date
- Sentiment Classification
- Sentiment Value

4 Methodology

The following steps were performed:

1. Data loading and preprocessing.
2. Date conversion and formatting.
3. Merging both datasets on the date column.
4. Handling missing values.
5. Exploratory Data Analysis (EDA).
6. Profitability and win rate analysis.
7. Visualization and interpretation of results.

5 Data Preprocessing

- Converted timestamps into datetime format.
- Extracted date from trade timestamps.
- Merged both datasets using the date field.
- Removed six rows containing missing sentiment values.

6 Results

6.1 Average Profit by Market Sentiment

Sentiment	Average Closed PnL
Extreme Fear	34.54
Extreme Greed	67.89
Fear	54.29
Greed	42.74
Neutral	34.31

Table 1: Average Profit by Sentiment

6.2 Win Rate by Market Sentiment

Sentiment	Win Rate (%)
Extreme Fear	37.06
Extreme Greed	46.49
Fear	42.08
Greed	38.48
Neutral	39.70

Table 2: Win Rate Analysis

6.3 Total Profit by Market Sentiment

Sentiment	Total Profit
Extreme Fear	739,110
Extreme Greed	2,715,171
Fear	3,357,155
Greed	2,150,129
Neutral	1,292,921

Table 3: Total Profit Analysis

7 Key Findings

1. Fear periods generated the highest total profit among all market sentiments.
2. Extreme Greed periods achieved the highest average profit per trade.
3. The highest win rate was observed during Extreme Greed conditions.
4. Neutral and Extreme Fear periods showed comparatively lower profitability.
5. Market sentiment significantly impacts trading performance.

8 Business Insights

- Sentiment indicators can improve trading decisions.
- Traders may increase exposure during high-confidence bullish periods.
- Risk management becomes essential during Extreme Fear periods.
- Combining sentiment indicators with technical analysis can improve profitability.

9 Conclusion

This study demonstrates a strong relationship between Bitcoin market sentiment and trader performance. Traders tend to achieve higher profitability and win rates during periods of Extreme Greed, while Fear periods contribute the highest cumulative profits. Therefore, incorporating market sentiment into trading strategies can lead to better decision-making and improved trading performance.

10 Future Work

Future research may include:

- Machine learning models for profit prediction.
- Sentiment forecasting using social media data.
- Portfolio optimization based on market sentiment.
- Strategy backtesting using multiple cryptocurrencies.

GitHub Repository

<https://github.com/kunalv2002/Bitcoin-Market-Sentiment-Analysis-Assignment>